



Shahjalal University of Science and Technology, Sylhet.

Lab Report

Department of Electrical and Electronic Engineering

Course Code: **EEE 336**

Course Title: **Control System I Lab**

Experiment No.: **05**

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Example M5.1

Take a typical underdamped second-order transfer function and import it to the LTI viewer window.

Also show the characteristics: Rise time, peak time, settling time and steady state final value.

Ans: An Example of a 2nd order system through MATLAB code, figure:

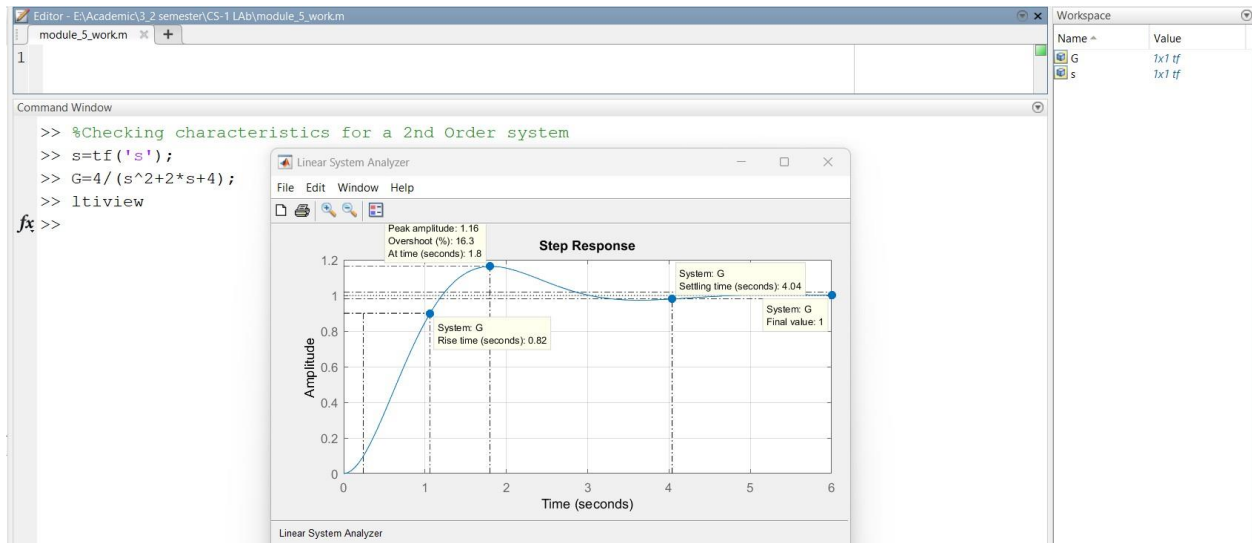


Fig: The characteristics

Consider the position control system shown in Fig. M5.8.

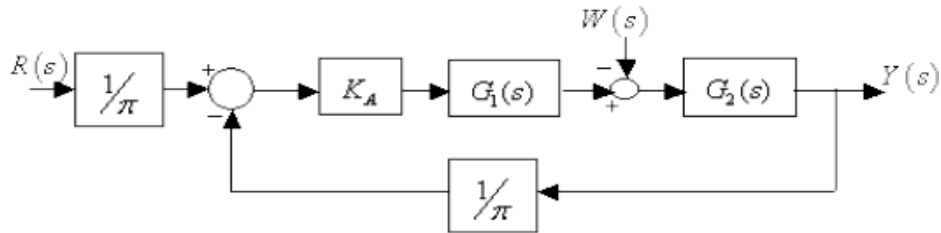


Fig. M5.8

(transfer function of motor, gears, and load)

$$G_1(s) = \frac{100}{s+100}$$

(transfer function of power amplifier)

K_A = preamplifier gain

$\frac{1}{\pi}$ = sensitivity of input and output potentiometers

Ans: MATLAB Code and figure of given block diagram:

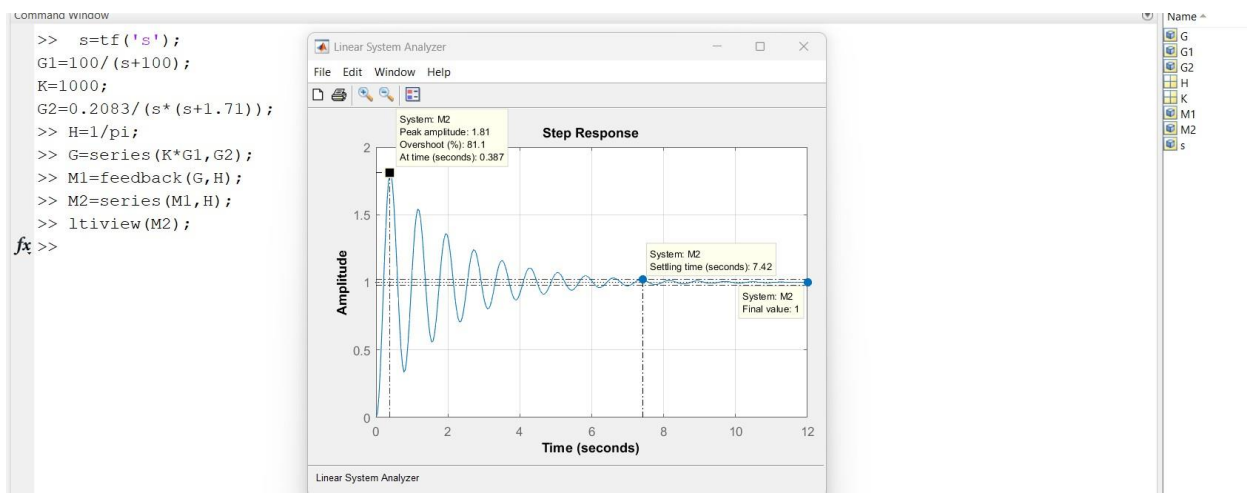


Fig: Code and Block Diagram

Example M5.2

Consider a unity feedback system shown in Fig. M5.11.

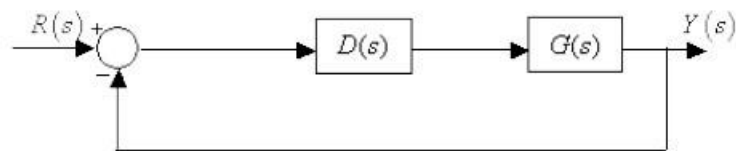


Fig. M5.11

$$G(s) = \frac{59.292}{s^2 + 6.9779s + 15.1239} \text{ (plant transfer function)}$$

$$D(s) = \left(K_P + \frac{K_I}{s} + K_D s \right) \text{ (controller transfer function)}$$

We will consider the following three controllers:

$$D1(s) = 3; D2(s) = 3 + (15/s); \text{ and } D3(s) = (3 + (15/s) + 0.3s)$$

Ans: MATLAB code and figure:

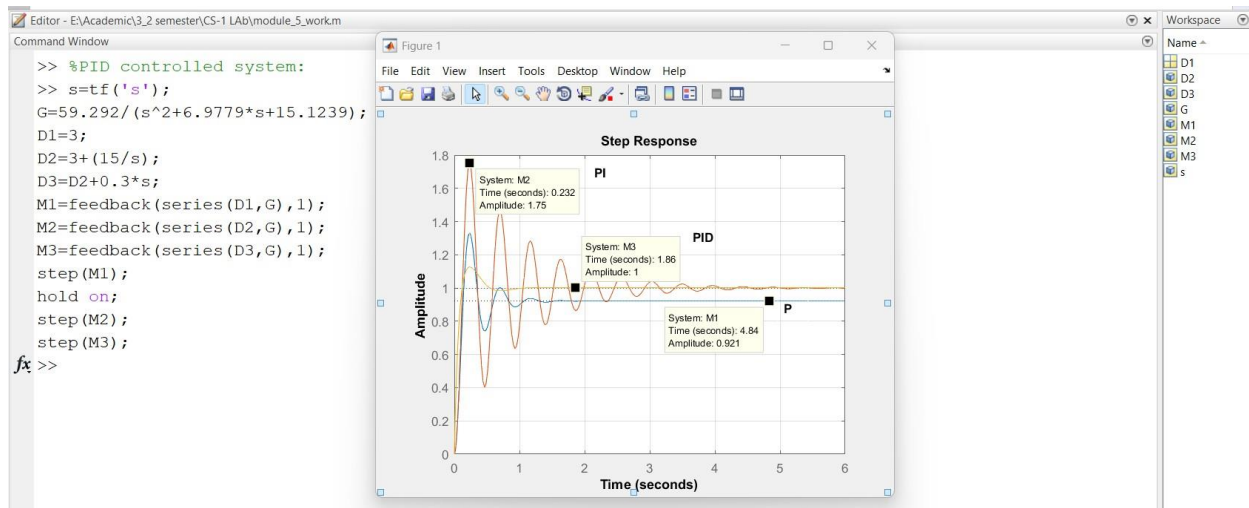


Fig: Step Response and Code

Example M5.3

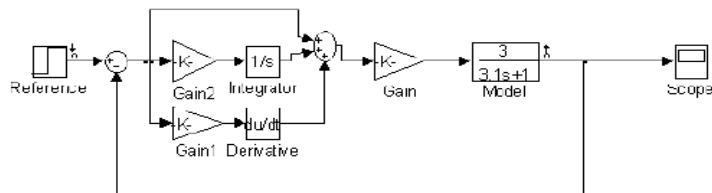
Consider a unity feedback, PID controlled system with following parameters:

Plant transfer function: $G(s) = \frac{3}{3.1s + 1}$

PID controller transfer function: $D(s) = 3.37 \left(1 + \frac{1}{1.2s} + 0.06s \right)$

Reference input: $R(s) = \frac{30}{s}$

A Simulink block diagram is shown in Fig. M5.13.



Ans: Simulink block diagram and LTI figure:

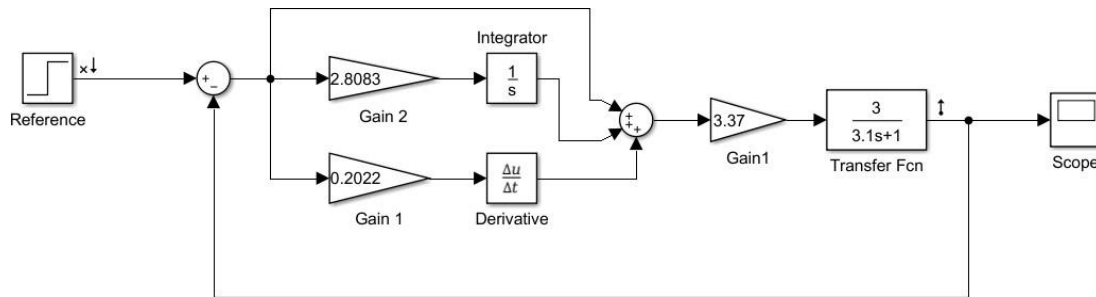


Fig: Simulink Model

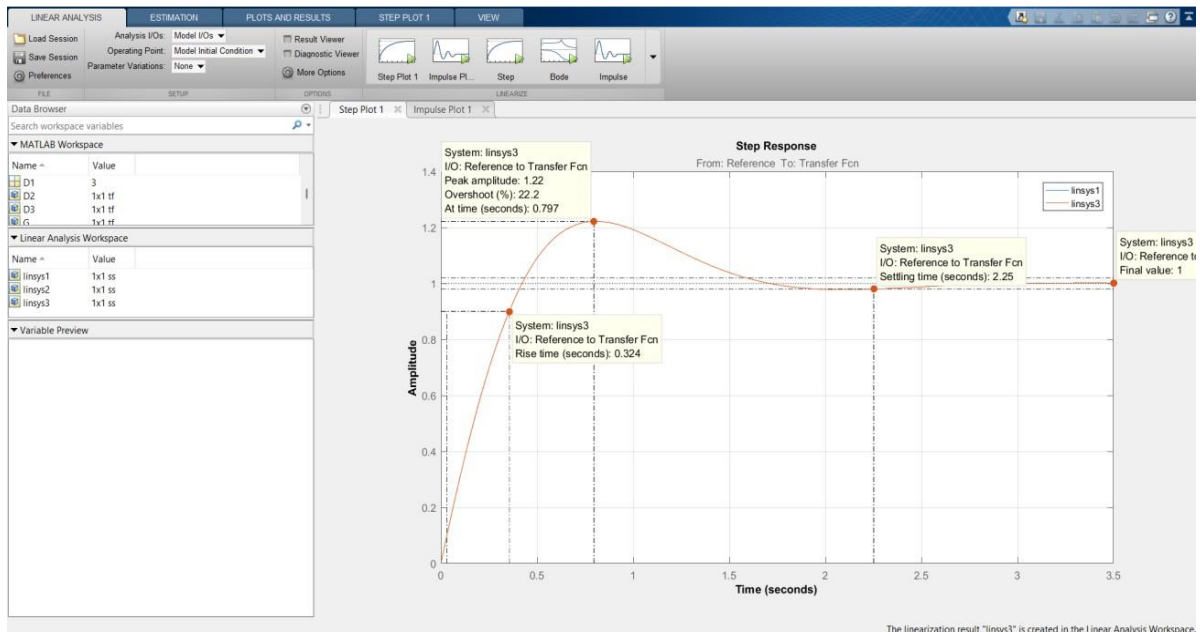


Fig: Step Response

Example M5.4

Consider a unity feedback, PI controlled system with following parameters:

Plant transfer function: $G(s) = \frac{8.55}{0.55s + 1}$

Plant dead-time: $\tau_D = 0.15$ minutes

PI controller transfer function: $D(s) = 0.09 + \frac{0.95}{s}$

Actuator saturation characteristic: Unit slope; maximum control signal =1.7.

Reference input: $R(s) = \frac{10}{s}$

A Simulink block diagram is shown in Fig. M5.18. PID controller block is available in **Additional Linear** block set in **Simulink Extras** block library. The parameters of PID controller block have been set as : $P = 0.09$, $I = 0.95$, and $D = 0$.

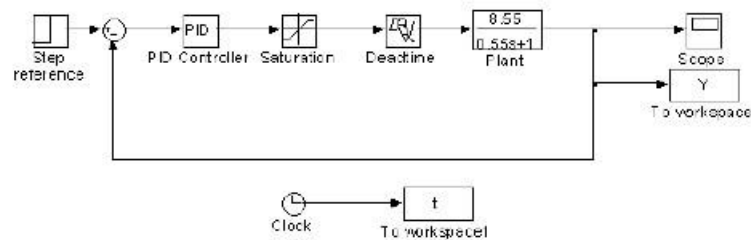


Fig. M5.18 [\(download\)](#)

Ans: Simulink block diagram and Matlab code:

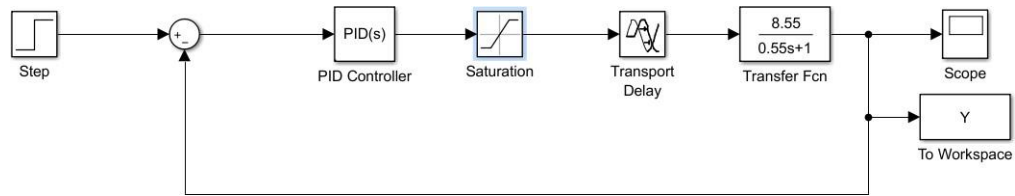


Fig: Simulink Model

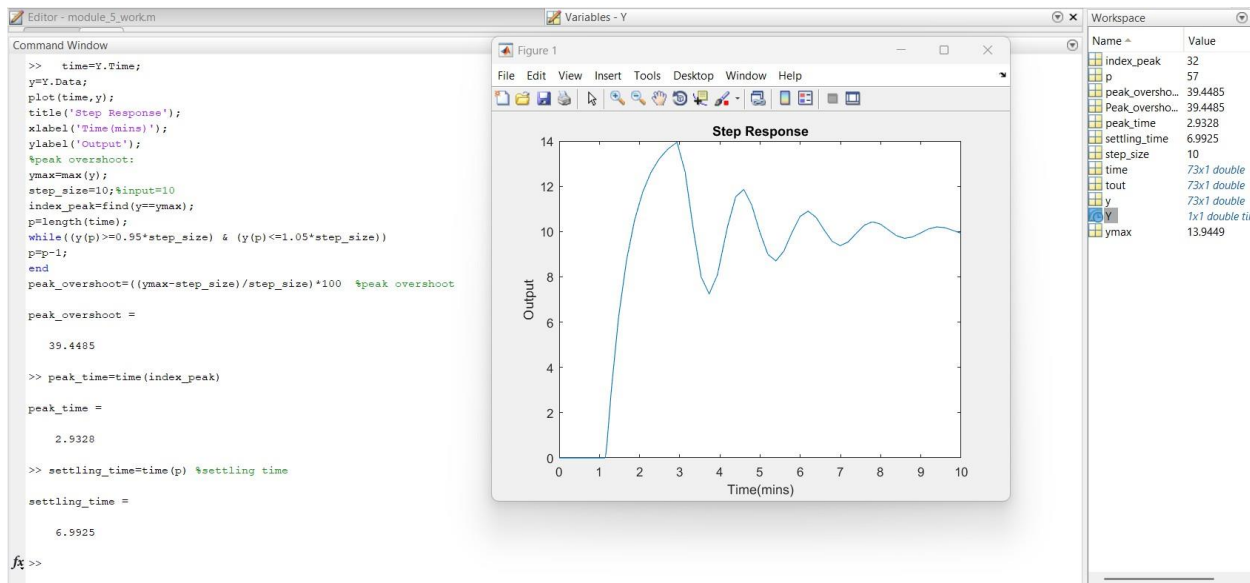


Fig: Step response and Code